

Exit Exam question 2015

1. Which parameter is used to assess and evaluate software architectures?
 - a. Responsiveness of Architectures
 - b. Durability of Architectures
 - c. Architectural quality in attribute
 - d. Number of components in the Architecture's
- 2.
3. What we use to pass the data between activities in Android?
 - a. Broadcast receiver
 - b. PostgreSQL Database
 - c. Intent
 - d. content provider
4. The following UML design shows student class. Assume that the Attributes have declared as private and the method as public. The get method return the value of the attribute and calculateAge() returns the student age. Which of the following would not raise an error when you run this program?

Student
Name: String
Department: String
Phone: String
Age: int
Student (String, String, String, int)
SetDept(String)
GetName(): String
GetDept():String
GetYearOfBirth():int
CalculateAge(): int

- a. Student st1 = newStudent("john", "SE", "0913222222", 20)
- b. st1 Student = newStudent("john", "CS", "02085332455", 23)
- c. Student st1 = newStudent("john", "IT", "2000",)
- d. Student st1 = newStudent("john", "CS", "0913222222", "25")

5. What will be the output of the following code snippet?

```
itex<script type="text/JavaScript" language = "JavaScript" >
var a = "exitexam";
var result = "a.substring(2,6)";
document.write(result);
```

- a. texam
- b. xite
- c. itex
- d. xitex

6. Which one of the following is wrong regarding to interface?

- a. Like a class, an interface is reference data type.
- b. Interface abstract method are accessed using interface instances.
- c. Interface includes abstract methods.
- d. One class can implement multiple interfaces.

7.

8. For the schema Hotel (Hotel_id:integer, Hotel_name: String, Sub_city: string) which SQL statement correctly inserts Tuples to Hotel relation. (Assume Hotel_id is primary key).

- a. Insert INTO HOTEL (Hotel_id, Hotel_name, Sub_city)VALUES ('Hilton','Yeka')
- b. Insert INTO HOTEL (Hotel_id, Hotel_name, Sub_city)VALUES (1,'Hilton','Yeka')
- c. Insert INTO HOTEL (1,'Hilton','Yeka')
- d. Insert Values INTO HOTEL(Hotel_id, Hotel_name, Sub_city)VALUES (1,'Hilton','Yeka')

9.

10. Which of the following shows the five stages in Tuckman's model of team development, in sequential order?

- a. forming, storming, performing, norming and adjourning
- b. norming ,forming, storming, performing and adjourning
- c. storming, forming, norming, performing and adjourning
- d. forming, storming, norming, performing and adjourning

11. Which Statement is correct about the difference between functional and nonfunctional requirement?

- a. Unlike functional requirement, nonfunctional requirement is stable.
- b. Both are the same most of the time and only vary under certain contexts.

- c. Functional requirements are decided by the customer while nonfunctional requirements are decided by developers
- d. Functional requirements are the service of the system provide while nonfunctional requirements are constraints on the service or functions offered by the system.

12.

13. If the estimate the total discounted benefits for a project is birr 120,000 and total discounted cost is birr 100,000 what is the estimated return on investment (ROI) percentage?

- a. 20%
- b. 12%
- c. 30%
- d. 10%

14. A queue in which the item most recently added is always the first one out refers to:

- a. LIFO queue
- b. FIFO queue
- c. Real time queue
- d. Priority queue

15.

16.

17. A firewall is used in a system connected to a wide area network to

- a. Fire spreading via network through cables
- b. Avoid Spreading of fire in the network
- c. Stop Unauthorized access by hackers
- d. Scanning for viruses in various files

18. The permission –rwxr-r- represented in octal expression will be:

- a. 766
- b. 700
- c. 744
- d. 755

19. Which of the following is the correct order of packet data unit as data moved in the OSI reference model from bottom to top?

- a. Segment, Packet, Frame, Bit
- b. Segment, Frame, Packet, Bit
- c. Bit, Frame, Packet, Segment

d. Bit, Packet, Frame, Segment

20. Which of the following statement is true regarding triple DES?

- a. 192-bit keys on 64-bit blocks of plain text.
- b. Uses 128-bit blocks of plain text and 92-bit keys and apply DES algorithm thrice.
- c. Work with 192-bit blocks of plain text and applies DES algorithm once.
- d. Functioning on 64-bit blocks of plain text and 56-bit keys by applying DES algorithm for three rounds.

21.

22. An HTTP request message always contains_____?

- a. a status line, a header, and a body
- b. a header only
- c. a header and a body
- d. a request line and a header

23. Which of the following is refers to unauthorized disclosure of information?

- a. authorization
- b. authentication
- c. integrity
- d. confidentiality

24. Anything that can perceive its environment through sensory and act as upon environment through effectors

- a. Agent
- b. Expert System
- c. Intelligence
- d. API

25. Assume that there is a pet family from which Pets like Dog and Cat share behavior from every Pet family has a name. However, the pet families have some specific behaviors that are exhibited by only particular pets. For instance, only Dogs can fetch something while both cats and dogs can speak which of the following object oriented design principle would be suitable for this case?

- a. Information hiding
- b. Encapsulation
- c. Polymorphism

d. Inheritance

26. Which of the following statement is not addressing the banker's algorithm?
- a. The banker's algorithm considers each request before it occurs.
 - b. It is an extension of the deadlock detection algorithm.
 - c. It is modeled on the way a small-town banker might deal with a group of customers to whom he/she has granted lines of credit
 - d. It is scheduling algorithm.
27. An informed search algorithm that provides exponential time complexity in the worst-case scenarios and guarantees optimality when searching a solution refers to:
- a. A* search
 - b. greedy best-first search
 - c. uniform cost search
 - d. Iterative deepening A*search
28. Which principles state that 80% of the problem can be fixed with 20% of the entire effort?
- a. Pareto principle
 - b. Parametric principle
 - c. Pairwise principle
 - d. Partition principle
29. Which one of the following is an appropriate sequence that depicts the model development process in data mining?
- a. Model training, Model Testing, Model Evaluation, Model Deployment
 - b. Model Evaluation, Model training, Model Testing, Model Deployment
 - c. Model training, Model Testing, Model Deployment, Model Evaluation
 - d. Model training, Model Evaluation, Model Testing, Model Deployment
30. Which method has the same name as that of its class?
- a. delete
 - b. class
 - c. constructor
 - d. finalize
31. Within OAuth, what component validates the user's identity?
- a. Authorization server
 - b. resource server
 - c. browser
 - d. client

32. The result of the following program after running will be

```
Class PrintResult {  
Public static void main (String arg [ ]) {  
int arr{ }={3,4,5,6,7};  
for(int i=0; i<=arr.length-2; ++i){  
system.out.println(arr[i]+ ",")  
}  
}}
```

- a. 3,4
- b. 3,4,5
- c. 3,4,5,6,7
- d. 3,4,5,6

33. While optimizing our relation, if we found that no multivalued attributes and partial dependencies exist in a relation then the relation is in what normal form?

- a. 1NF
- b. 3NF
- c. 4NF
- d. 2NF

34. A Techniques for generating plans with conditionals and loops are almost identical to those for generating programs from logical specification is called:

- a. Automatic programming
- b. Automatic monitoring
- c. Automatic recursive
- d. Automatic learning

35. Which one of the following is the correct identifier in c++?

- a. 7variable
- b. 7VARIABLE
- c. \$variable
- d. variable_1234

36. Which of the following is true about the abstract class?

- a. cannot be inherited
- b. a mechanism of encapsulation
- c. cannot be instantiated
- d. can contain both abstract and non-abstract methods

37. In software risk management, accepting risk happening but do not do anything is known as
- a. risk transfer
 - b. risk retention**
 - c. risk avoidance
 - d. risk reduction
38. When a process is in a “Blocked” state waiting for some input-output services and after the service is completed it will go to a state known as:
- a. Suspended
 - b. Terminated
 - c. Running
 - d. Ready**
39. “The system architecture should be designed using fine-grain, self-contained components. Producers of data should be separated from consumers and shared data structures should be avoided.” For which requirement this architecture description applies?
- a. Availability
 - b. maintainability**
 - c. performance
 - d. security
40. Which slogan is against agile method philosophy?
- a. Following a plan over responding to changes**
 - b. Individuals and interactions over processes and tools
 - c. Customer collaboration over contract negotiation
 - d. working software over comprehensive documentation
41. APK stands for
- a. Android Package Kit**
 - b. Android Platform Kit
 - c. Android Phone Kit
 - d. Android Page Kit
42. Which statement describes the main goal of software testing as a part of the quality assurance process?
- a. to achieve project timelines and deadlines
 - b. to generate software requirement and design documentation
 - c. to guarantee that the software meets the specified requirement**
 - d. to monitor the entire SDLC

43. The correct structure of the for loop statement is:
- a. for(initialization; condition)
 - b. for(increment/decrement; initialization; condition)
 - c. for(condition, initialization, increment/decrement)
 - d. **for(initialization; condition; increment/decrement)**
44. Which of the following is not true regarding algorithm development during problem solving?
- a. should be developed by considering the platforms on which it runs
 - b. Is step wise logical description of how to solve the problem
 - c. **Is developing considering the details of the programming language**
 - d. Is equivalent to the programming language code
45. What would be the most likely consequence if every software stakeholders were rushing to be approved for their requirement without discussing with other stakeholders?
- a. Attains the project schedule
 - b. Results in excess requirements
 - c. **leads to conflict requirement**
 - d. Requirement become unambiguous
46. Which of the following is true regarding model development in machine learning?
- a. Much focus should be given to test data than training data
 - b. **Training with large dataset boosts model performance**
 - c. Test set and Training set should have fifty-fifty overlap
 - d. **proportion of data set for training should be less than the test set**
47. All layout classes are subclasses of which of the following?
- a. android.view.RelativeLayout
 - b. android.view.Layout
 - c. **android.view.ViewGroup**
 - d. android.view.Widget
48. What will be the output for the following code if you enter 4 and 3 for b and e respectively
- ```
#include<iostream>
Using namespace std;
int main(){
int b, e, r=1;
```



```

count<<"enter b and e";
cin>>b>>e;
for (int i=1; i<=e; i++){
f=r*b;
}
cout<<f;
return 0;
}

```

- a. 81
- b. 12
- c. 64
- d. 256

49. Which of the following is not a valid host address within the network 172.17.128.0/21?

- a. 172.17.128.1/21
- b. 172.17.135.255/21
- c. 172.17.128.255/21
- d. 172.17.135.0/21

50. Which of the following is the correct sequence that will be followed during software evolution process?

- a. change request, release planning, change implementation, impact analysis, system release
- b. impact analysis, change request, change implementation, release planning, system release
- c. change request, impact analysis, change implementation, release planning, system release
- d. change request, impact analysis, release planning, change implementation, system release

51. Identify the lowest layer of android architecture

- a. Linux kernel
- b. Database
- c. Application Framework
- d. Application

52. Which of the following is true regarding the use of switches and hubs for network connectivity in the network?
- a. Switches do not forward broadcasts
  - b. Switches increase the number of collision domains in the network
  - c. Switches take less time to process frames than hubs take
  - d. Hubs can filter frames
53. Select the one that is not correct related to generalization in system modeling.
- a. In generalization, the attribute and operation associated with the highest-level classes are also associated with lowest level classes.
  - b. Common information will be maintained in one place
  - c. Facilitate easy modification of data
  - d. The higher level classes are more specific than the lower-level classes by adding specific attributes and operation
54. A of process management which is waiting for a resource that is owned by another process refers to:
- a. deadlock
  - b. preemption
  - c. overloading
  - d. queueing
55. A set of processes management which is waiting for a resource that I owned by another process refers to:
- a. Deadlock
  - b. Preemption
  - c. Overloading
  - d. Queuing
56. “Database is a logically coherent collection of data with some inherent meaning “ describes that
- a. All attributes of the entity should be related
  - b. there should be one primary key attribute
  - c. the entity should be physical object
  - d. number of attributes should be limited
57. Which of the following is a resource optimization techniques in which start and finish dates are adjusted based on resource constraints with the goal of balancing the demand resources with the available supply?
- a. Responsibility assignment matrix

- b. Resource smoothing
- c. Resource leveling
- d. Resource grouping

58. Business people always remodel their business by closely studying their customers purchasing habits over the course of their purchase history. Accordingly, a salesperson at Markato shop noticed that people who always buy shoes will also buy socks with probability of 0.75 and hence decided to put shoes and socks side by side for ease of accessibility. Which big data analytics technique could this salesperson could employ?

- a. Regression Modeling
- b. Clustering algorithm
- c. Predictive modeling
- d. Association rule

59. Given the following layered architecture which ordering relation is correct and didn't encountered any architectural erosion? Assume there are four layers: A,B,C and D

|   |
|---|
| A |
| B |
| C |
| D |

- a. (A,B)
- b. (D,A)
- c. (A,C)
- d. (C,B)

60. Which of the following is among benefits provided with access control lists (ACLS) implementation for software security based application?

- a. ACLs monitor the number of bytes and packets
- b. ACLs provide high network availability
- c. Virus detection
- d. ACLs classify and organize network traffic

61. Default method while Submitting a form is

- a. Post method
- b. Get Method
- c. Put Method
- d. Set Method

62. The process that generates activities such as project schedule, cost estimations and work breakdown structures is described as:

- a. Project Initiating
- b. Project Executing
- c. Project Closing
- d. Project Planning

63. Given the following fragment of code, how many tests are required for 100% decision coverage?

```
 if width > length
 then biggest_dimension = width
 if height > width
 then biggest_dimension = height
 end_if
else biggest_dimension = length
if height > length
then biggest_dimension = height
end_if
end_if
```

- a. 6
- b. 4
- c. 2
- d. 1

64. How can we select an element with a specific class in CSS?

- a. ^
- b. -
- c. #
- d. .

65. Select an activity that suits in the fundamental test process which includes evaluation of the testability of the requirement and system?

- a. Test analysis and design
- b. Test analysis and requirements
- c. Test analysis and implementation
- d. Test analysis and planning

66. Which of the following is an appropriate sequence of database design processes?

- a. Logical design database, Enterprise data modeling, Physical database design, Database implementation
- b. Enterprise data modeling, Logical design database, Database implementation, Physical database design
- c. Enterprise data modeling, Logical design database, Physical database design, Database implementation
- d. Logical design database, Enterprise data modeling, Physical database design, Database implementation

67. Which of the following is not among the purposes of software testing?

- a. Identify shortcomings
- b. Improving product acceptance
- c. Enhancing reliability
- d. Requesting more design and implementation time

68. Which software process model will you use if you want to deliver different functionalities (modules) of the software product that have different priority at different time?

- a. Incremental model
- b. Spiral model
- c. Waterfall model
- d. Linear model

69. What is the time complexity for the following code snippets?

```
function someFunction(n)
 for(var i=0; i<n*10; i++){
 console.log(n);
 }
```

- a.  $O(n^2)$
- b.  $O(\log n)$
- c.  $O(\log^2 n)$
- d.  $O(n)$

70. Select the wrong statement regarding traceability in software development projects

- a. Traceability shows the dependency among two or more requirements if any

- b. Requirements should be linked to the respective stakeholders who generated them
  - c. Every design element (component) should be linked back to the requirement
  - d. Traceability traces the overall cost and schedule of every activity in the requirements
71. Which activity comes last in fundamental program development in c++?
- a. Memory allocation for variables
  - b. Linking
  - c. Compiling program
  - d. Writing program
72. Which design pattern is used to restore the state of an object to its previous state?
- a. Visitor pattern
  - b. Momento pattern
  - c. Observer pattern
  - d. Iterator pattern
73. Which of the following describes a set of standards and an associated network protocol that allow establishing a secure channel between a local and a remote computer?
- a. Secure socket layer(SSL)
  - b. Secure shell(SSH)
  - c. Authentication, Authorization and Accounting(AAA)
  - d. Simple network management protocol(SNMP)
74. What components hides the distinctions or boundaries between various micro services from end client application?
- a. A Layered system
  - b. API gateway
  - c. API proxy
  - d. API logging
75. Select a fundamental issue that may not be considered by the software architects during the architectural design process?
- a. Which architectural organization is best for delivering the functional requirements
  - b. How to decompose structural components in to sub components

- c. How the system will be distributed across a number of cores or processes
- d. Which architectural patterns or styles to use

76. Which of the following statement is not correct statement regarding of project life cycle?

- a. In agile life cycle the project scope is generally determined early in the project life cycle, but time and cost estimates are routinely modified
- b. In waterfall life cycle, the project scope, time and cost are determined in the early phase of the life cycle.
- c. In agile life cycles, the project scope cannot be outlined and agreed before the start of iteration.
- d. In an incremental life cycle, the deliverable is produced through a series of iterations that successively add functionality within a predetermined time frame.

77. Which of the following is not among the characterizing features of database approach?

- a. Application-data dependency
- b. Sharing of data
- c. Self-describing
- d. Data abstraction

78. What is an activity in android?

- a. android class used to configure the android application
- b. It is an intent
- c. android package file holding all the package used
- d. A single screen in an application with supporting java code

79. Which of the following is not an approach used by IT security specialists to enhance the security level of the network?

- a. Use of Intrusion detection system
- b. Use of Wana cry protocol
- c. Use of physical security
- d. Use of intrusion prevention system

80. As a software tester, when do you implement configuration management procedures?

- a. During test planning
- b. During test closing
- c. During test execution

- d. During test initiation
81. Identify the one that comes first in data mining process
- a. Data cleaning
  - b. Data integration
  - c. Business understanding
  - d. Data selection
82. A TCP/IP layer which is responsible for addressing, path selection and routing refers:
- a. Application layer
  - b. Internet layer
  - c. Transport layer
  - d. Network Access layer
83. Which of the following analysis mechanism defines the input for which the algorithm takes the least time (fastest time to complete)?
- a. Worst case
  - b. Best case
  - c. Standard case
  - d. Average case
84. Which learning algorithm is usually applied to data that does not contain any label information, so we do not know what the right output should be?
- a. Transfer learning
  - b. Supervised learning
  - c. Unsupervised learning
  - d. Reinforcement learning
85. Depending on the organization of the elements which of the following approach is a non-linear data structure?
- a. Queue
  - b. Stack
  - c. Tree
  - d. Linked list
86. Which one of the following can be among possible source of unstructured data?
- a. Apple employee database
  - b. RDBMS systems
  - c. Registrar student database
  - d. Google search engine



87. Select the one that is not related with the purpose of properly documenting software architecture
- a. For Stakeholder communication
  - b. To easily figure out the source code flow
  - c. For critical system design decisions
  - d. To know how the system is organized and interoperate
88. One of the following statement is correct about Artificial Intelligence (AI)?
- a. It refers to the development of machine that can only perform physical tasks.
  - b. It refers to the development of machine that can only perform simple, repetitive tasks.
  - c. It refers to the development of machine that can perform tasks only in a controlled laboratory setting.
  - d. It refers to the development of machine that can perform tasks that typically require human intelligence.
89. Which of the following is a process of selecting program paths in such a manner that certain branches (i.e, outgoing edges of nodes) of a control flow graph are covered by the execution of those paths?
- a. Branch coverage
  - b. Branch expansion
  - c. Branch control
  - d. Branch boundary
90. Give a situation two hosts attempt on half-duplex Ethernet LAN to send data concurrently, resulting in a collision and subsequent to this collision, what will the hosts do?
- a. The jam signal indicates that the collision has been cleared
  - b. An electrical pulse indicates that the collision has cleared
  - c. The hosts will attempt to resume transmission after a time delay has expired
  - d. The router on the segment will signal that the collision has cleared
91. What is the time complexity of below function?

```
Public void function (int n){
 for(int i=0; i<n; i++)
 for(int j=0; j<i*i; j++)
 if(j%i==0){
 for(int k=0; k<j; k++)
```

```
 print("**");
 }
}
```

- a.  $O(n^2)$
  - b.  $O(n \log n)$
  - c.  $O(n^7)$
  - d.  $O(n^5)$
92. Identify the design principle that does not apply to software system
- a. design should be traceable to the analysis model
  - b. design should be structured to accommodate change
  - c. design should exhibit uniformity and integration
  - d. design should be reinventing the wheel from scratch
93. You are required to write a program that iteratively takes 50 numbers from user input and adds them up only if the numbers are positive and skips if not. Which control structure best fits this scenario?
- a. break
  - b. continue
  - c. for
  - d. jump
94. Which of the following UML elements is wrongly applied?
- a. Class diagram shows the object classes in a system and their relation
  - b. Activity diagram shows how the system reacts to internal and external events
  - c. Use case diagram represents an interaction with a system
  - d. Sequence diagram shows the sequence of interactions required to complete some operation
95. A memory management policy that that decisions must be made as to which page or pages are to be supplanted when a memory is full is known as:
- a. Cleaning policy
  - b. Load policy
  - c. Replacement policy
  - d. Fetch policy
96. Which software model would you apply for business operating in a changing environment where the initial requirements in inevitably change due to some internal and external factors?
- a. Agile development method

- b. Spiral model
- c. Water fall model
- d. Code and run

97. Assume that you are assigned to design a dormitory management database. A dormitory is located on some building and has attributes specified as DORM (DormId, floornumber) and a DormId is primary key based on university's rule, six students can be assigned to one dorm. Students can be tracked with attributes STUDENT (IdNo, Name, Departement) and IdNo is a primary key. Based on this scenario description, which of the following SQL statement is correct to establish relationship between DORM and STUDENT assuming DorId is foreign key?

- a. FOREIGN KEY (DorID) REFERENCES DORM(DormID)
- b. STUDENT table is a reference relation
- c. FOREIGN KEY (DorID) REFERENCED by DORM(DormID)
- d. DORM table is a referencing relation

98. Which one of the following statements is wrong regarding try...catch construct?

- a. When exception occurs, the execution of try block is interrupted
- b. The execution control is moved to catch block after exception occur in try block.
- c. The catch block is used to handle exceptions that occur in try block.
- d. The catch block is executed once regardless of exceptions in try block

99. Which of the following is not the function of Cross-over UTP cable?

- a. To connect switch to switch
- b. To connect pc to pc
- c. To connect router to pc
- d. To connect switch to pc

100. Who is responsible for documenting faults found during the software development process?

- a. Tester
- b. Developer
- c. Scrum master
- d. Requirement engineer

